

Web Programming (CS4032) / Sessional-II

Web Engineering (SE3003) Exam

Date: April 8, 2026

Course Instructor(s)

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Total Time: 1 hour

Total Marks: 40

Total Questions: 3

Roll No

Section

Student Signature

Do not write below this line

Attempt all questions on the answer sheet

CLO2: Learn basics of client and server-side programming

[15 marks]

Question 1

Pakistan ▼

Lahore ▼

Create React **Function** Components to display a list of countries and cities as dropdowns. Lists can be loaded from `/countries` and `/cities/<country_name>` end points/urls in the form of comma-separated values or JSON array using `fetch` API. For instance, list of countries can be as follows:

`['Pakistan','India','Bangladesh','SriLanka']`

Likewise, list of cities for Pakistan can be:

`['Karachi', 'Lahore', 'Peshawar', 'Quetta', 'Islamabad']`

Selecting an appropriate country in the respective dropdown shall fetch and load corresponding city dropdown. Following HTML can be used to specify a dropdown:

```
<select>
  <option>1</option>
  <option>2</option>
  <option>3</option>
</select>
```

CLO2: Learn basics of client and server-side programming

Question 2

[15 marks]

Task name :

Priority :

Due date : ☐

Assigned To :

Description

Consider a form illustrated above to store and assign a task to a resource in a task management system:

- Design schema to store task and assignee information separately in **Mongo DB**
- Write code to handle request using **Node / Express**, insert the information in database and return a proper response.

Note: Form will be submitted using a **POST** request. “Assigned To” dropdown shows employee / resource **name** on the form. Make other necessary assumptions about input field names and selected values. There is no need to write any HTML / JSX for the form. Simply write the server-side code.

CLO3: Learn modern practices along with prevalent technologies and frameworks

Question 3

[5+5 marks]

Answer the following questions:

1. Describe state management techniques in a React / Redux application
2. Describe state management approaches for server-side programming